

Heteropar Notebook

Source: C:\Users\Public\New project\riscv-rvv-ime-gemm-microkernels\Heteropar.ipynb

Code cell 1

```
import os
import numpy as np
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import matplotlib.patheffects as pe
import seaborn as sns
from matplotlib.patches import Patch
from IPython.display import Image, display

# #####
# Springer / publication style
# #####
sns.set_theme(style="whitegrid", context="paper")

matplotlib.rcParams.update({
    "pdf.fonttype": 42,
    "ps.fonttype": 42,
    "font.family": "serif",
    "font.serif": ["Times New Roman", "Times", "DejaVu Serif"],
    "mathtext.fontset": "stix",

    "axes.linewidth": 0.90,
    "axes.edgecolor": "black",
    "axes.spines.top": False,
    "axes.spines.right": False,

    "xtick.major.width": 0.80,
    "ytick.major.width": 0.80,
    "xtick.major.size": 3.6,
    "ytick.major.size": 3.6,
    "xtick.direction": "out",
    "ytick.direction": "out",
})

# #####
# Output
# #####
OUTPUT_DIR = "output"
os.makedirs(OUTPUT_DIR, exist_ok=True)

SAVE_PATH_PNG = os.path.join(OUTPUT_DIR, "fp_64_8x4_springer_final_largerfont.png")
SAVE_PATH_PDF = os.path.join(OUTPUT_DIR, "fp_64_8x4_springer_final_largerfont.pdf")

# #####
# Data
# #####
DATA = [
    # FP64
    ("FP64", "2", 1, 3.613117, 0.392615),
    ("FP64", "2", 2, 3.464250, 0.412135),
    ("FP64", "2", 4, 3.316667, 0.407032),
    ("FP64", "2", 8, 3.421700, 0.367016),
    ("FP64", "4", 1, 2.504333, 0.219997),
    ("FP64", "4", 2, 2.717317, 0.163361),
    ("FP64", "4", 4, 2.630733, 0.194312),
    ("FP64", "4", 8, 2.640033, 0.205592),
    ("FP64", "8", 1, 0.640517, 0.000474),
    ("FP64", "8", 2, 0.554383, 0.003143),
    ("FP64", "8", 4, 0.550567, 0.000259),
    ("FP64", "8", 8, 0.615300, 0.000792),

    # INT8
    ("INT8", "mf8", 1, 3.875800, 0.040295),
    ("INT8", "mf8", 2, 4.375483, 0.024772),
    ("INT8", "mf8", 4, 4.134017, 0.023203),
    ("INT8", "mf8", 8, 4.023583, 0.026163),
    ("INT8", "mf4", 1, 3.889300, 0.023601),
    ("INT8", "mf4", 2, 4.358783, 0.033337),
    ("INT8", "mf4", 4, 4.144000, 0.027036),
    ("INT8", "mf4", 8, 4.047717, 0.029463),
    ("INT8", "mf2", 1, 3.859500, 0.064757),
```



```

    "INT8": "GOPS",
    "IME": "GOPS",
}

UNROLL_LABEL = {
    1: "U1",
    2: "U2",
    4: "U4",
    8: "U8",
}

Y_LIMITS = {
    "FP64": (0.0, 5.05),
    "INT8": (3.72, 4.62),
    "IME": (14.0, 22.7),
}

BAR_WIDTH = {
    "FP64": 0.115,
    "INT8": 0.078,
    "IME": 0.125,
}

GROUP_WIDTH = {
    "FP64": 0.56,
    "INT8": 0.74,
    "IME": 0.32,
}

LABEL_FONT = {
    "FP64": 8.0,
    "INT8": 8.0,
    "IME": 8.0,
}

CAPSIZE = 2.0

LABEL_Y_OFFSET = {
    "FP64": 0.070,
    "INT8": 0.055,
    "IME": 0.24,
}

# =====
# Helpers
# =====
def present_lmuls_for_family(fam):
    present = set(df.loc[df["Family"] == fam, "LMUL"])
    return [l for l in LMUL_ORDER if l in present]

def add_label(ax, x, y, text, fontsize):
    label = ax.text(
        x,
        y,
        text,
        ha="center",
        va="bottom",
        rotation=90,
        fontsize=fontsize,
        fontweight="bold",
        color="#111111",
        zorder=9,
        clip_on=False,
    )
    label.set_path_effects([
        pe.Stroke(linewidth=1.6, foreground="white"),
        pe.Normal(),
    ])
    return label

# =====
# Figure
# =====
fig, axes = plt.subplots(
    1, 3,
    figsize=(6.8, 4.85),
    dpi=300,
    gridspec_kw={
        "width_ratios": [1.15, 1.95, 1.15],
        "wspace": 0.34,
    }
)

```

```
fig.suptitle(
    r"Throughput Tuning of FP64 OpenBLAS 8 $\times$ 4, INT8, and IME for 10243 GEMM",
    fontsize=10.5,
    fontweight="bold",
    y=0.970,
)
```

```
for ax, fam in zip(axes, FAMILIES):
    sub = df[df["Family"] == fam].copy()
    lmul = present_lmul_for_family(fam)

    x = np.arange(len(UNROLLS))
    total_width = GROUP_WIDTH[fam]
    bar_width = BAR_WIDTH[fam]

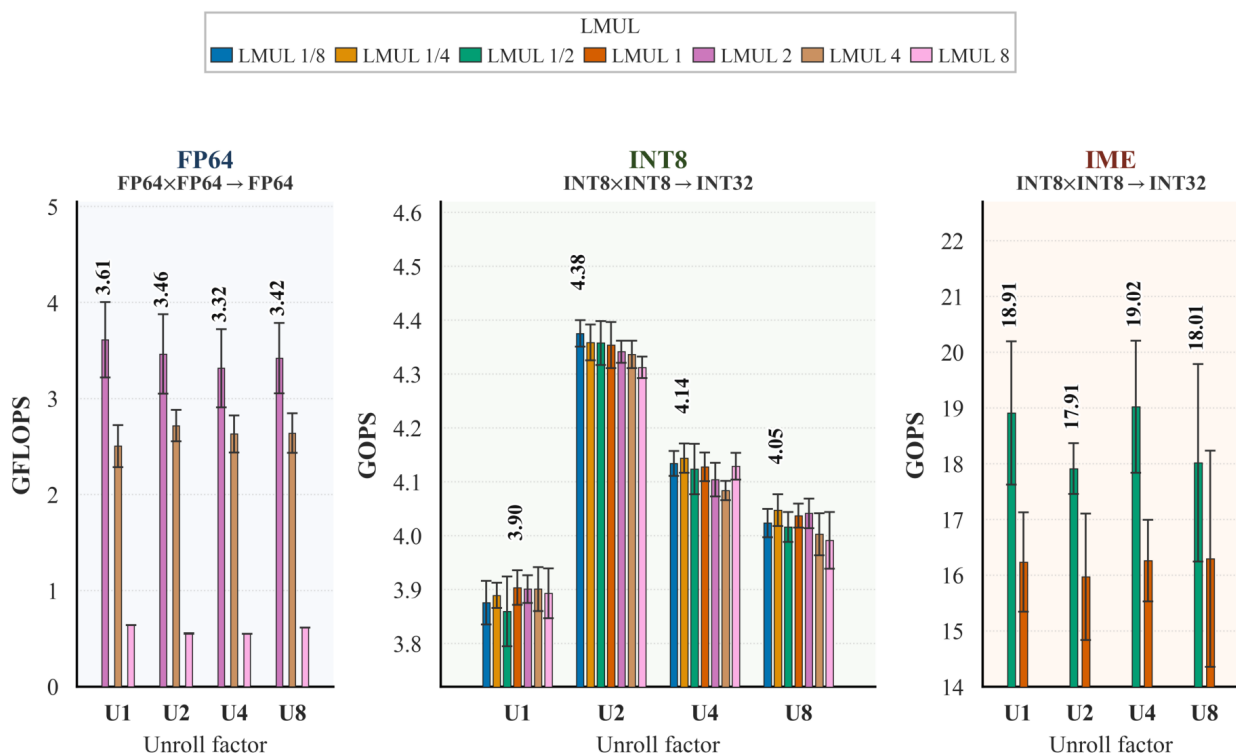
    offsets = np.linspace(
        -total_width / 2 + bar_width / 2,
        total_width / 2 - bar_width / 2,
        len(lmul)
    )

    ax.set_facecolor(FAMILY_FACE[fam])
    bar_info = {}

    for offset, lmul in zip(offsets, lmul):
        s = sub[sub["LMUL"] == lmul].set_index("Unroll").reindex(UNROLLS)
        va
... [output truncated for PDF readability]
```

PNG: C:\Users\HP 850 G7\output\fp_64_8x4_springer_final_largerfont.png
 PDF: C:\Users\HP 850 G7\output\fp_64_8x4_springer_final_largerfont.pdf

Throughput Tuning of FP64 OpenBLAS 8 $\times$ 4, INT8, and IME for 1024³ GEMM



<Figure size 2100x1275 with 1 Axes>

Throughput Tuning of FP64 OpenBLAS 8 $\times$ 4, INT8, and IME for 1024³ GEMM

Code cell 3

```
import os
import numpy as np
import pandas as pd
import matplotlib
```

```

import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import matplotlib.patheffects as pe
from matplotlib.patches import Patch

plt.close("all")

matplotlib.rcParams.update({
    "pdf.fonttype": 42,
    "ps.fonttype": 42,
    "font.family": "serif",
    "font.serif": ["Times New Roman", "Times", "DejaVu Serif"],
    "mathtext.fontset": "stix",
    "axes.linewidth": 0.90,
    "xtick.major.width": 0.80,
    "ytick.major.width": 0.80,
    "xtick.minor.width": 0.60,
    "ytick.minor.width": 0.60,
    "xtick.major.size": 3.6,
    "ytick.major.size": 3.6,
    "xtick.direction": "out",
    "ytick.direction": "out",
    "axes.spines.top": False,
    "axes.spines.right": False,
    "figure.facecolor": "white",
    "savefig.facecolor": "white",
})

OUTPUT_DIR = "output"
os.makedirs(OUTPUT_DIR, exist_ok=True)

SAVE_PATH_PNG = os.path.join(
    OUTPUT_DIR,
    "ime_acceleration_scalar_fallback_8x4_1024.png"
)
SAVE_PATH_PDF = os.path.join(
    OUTPUT_DIR,
    "ime_acceleration_scalar_fallback_8x4_1024.pdf"
)

FIG_W, FIG_H = 6.8, 4.85
BAR_WIDTH = 0.16
GROUP_WIDTH = 0.34
CAPSIZE = 2.0

DATA = [
    ("IME-Capable Core Cluster", "1", 1, 16.6407, 0.6052),
    ("IME-Capable Core Cluster", "1", 2, 16.6404, 0.7303),
    ("IME-Capable Core Cluster", "1", 4, 17.2594, 0.9416),
    ("IME-Capable Core Cluster", "1", 8, 15.9583, 1.3444),

    ("IME-Capable Core Cluster", "mf2", 1, 18.6883, 1.0597),
    ("IME-Capable Core Cluster", "mf2", 2, 19.0822, 1.1558),
    ("IME-Capable Core Cluster", "mf2", 4, 19.7634, 0.9200),
    ("IME-Capable Core Cluster", "mf2", 8, 18.4307, 2.1154),

    ("Non-IME Core Cluster", "1", 1, 0.0303, 0.0200),
    ("Non-IME Core Cluster", "1", 2, 0.0362, 0.0205),
    ("Non-IME Core Cluster", "1", 4, 0.0236, 0.0158),
    ("Non-IME Core Cluster", "1", 8, 0.0333, 0.0194),

    ("Non-IME Core Cluster", "mf2", 1, 0.0437, 0.0202),
    ("Non-IME Core Cluster", "mf2", 2, 0.0364, 0.0204),
    ("Non-IME Core Cluster", "mf2", 4, 0.0351, 0.0196),
    ("Non-IME Core Cluster", "mf2", 8, 0.0299, 0.0180),
]

df = pd.DataFrame(DATA, columns=["Family", "LMUL", "Unroll", "Mean", "Std"])

FAMILIES = [
    "IME-Capable Core Cluster",
    "Non-IME Core Cluster",
]

UNROLLS = [1, 2, 4, 8]
LMUL_ORDER = ["mf2", "1"]

LMUL_LABEL = {
    "mf2": "LMUL 1/2",
    "1": "LMUL 1",
}

```

```

LMUL_COLORS = {
    "mf2": "#009E73",
    "1": "#D55E00",
}

FAMILY_FACE = {
    "IME-Capable Core Cluster": "#FFF8F2",
    "Non-IME Core Cluster": "#F7FAF6",
}

FAMILY_ACCENT = {
    "IME-Capable Core Cluster": "#944727",
    "Non-IME Core Cluster": "#41642F",
}

FAMILY_SUBTITLE = {
    "IME-Capable Core Cluster": "Cores 0--3, IME path",
    "Non-IME Core Cluster": "Cores 4--7, scalar fallback",
}

UNROLL_LABEL = {
    1: "U1",
    2: "U2",
    4: "U4",
    8: "U8",
}

Y_LIMITS = {
    "IME-Capable Core Cluster": (0.0, 22.8),
    "Non-IME Core Cluster": (0.0, 0.085),
}

Y_TICKS = {
    "IME-Capable Core Cluster": np.arange(0, 22.6, 2.5),
    "Non-IME Core Cluster": np.arange(0, 0.081, 0.01),
}

LABEL_FONT = {
    "IME-Capable Core Cluster": 8.0,
    "Non-IME Core Cluster": 7.6,
}

LABEL_Y_OFFSET = {
    "IME-Capable Core Cluster": 0.42,
    "Non-IME Core Cluster": 0.0022,
}

VALUE_FORMAT = {
    "IME-Capable Core Cluster": "{:.2f}",
    "Non-IME Core Cluster": "{:.3f}",
}

def add_label(ax, x, y, text, fontsize):
    label = ax.text(
        x,
        y,
        text,
        ha="center",
        va="bottom",
        rotation=90,
        fontsize=fontsize,
        fontweight="bold",
        color="#111111",
        zorder=8,
        clip_on=False,
    )
    label.set_path_effects([
        pe.Stroke(linewidth=1.60, foreground="white"),
        pe.Normal(),
    ])
    return label

fig, axes = plt.subplots(
    1, 2,
    figsize=(FIG_W, FIG_H),
    dpi=300,
    constrained_layout=False,
    gridspec_kw={
        "wspace": 0.28,
        "width_ratios": [1, 1],
    },

```

```

    },
)

fig.patch.set_facecolor("white")

fig.suptitle(
    r"INT8 GEMM: IME Acceleration and Scalar Fallback"
    "\n"
    r"$1024^3$ 8$\times$4 Kernel Across Heterogeneous RISC-V Cores",
    fontsize=10.0,
    fontweight="bold",
    y=0.975,
)

for ax, fam in zip(axes, FAMILIES):
    sub = df[df["Family"] == fam].copy()
    lmul = [l for l in LMUL_ORDER if l in set(sub["LMUL"])]

    x = np.arange(len(UNROLLS))
    offsets = np.linspace(-GROUP_WIDTH / 2, GROUP_WIDTH / 2, len(lmul))

    ax.set_facecolor(FAMILY_FACE[fam])
    bar_info = {}

    for offset, lmul in zip(offsets, lmul):
        s = sub[sub["LMUL"] == lmul].set_index("Unroll").reindex(UNROLLS)
        vals = s["Mean"].values
        errs = s["Std"].values
        xpos = x + offset

        ax.bar(
            xpos,
            vals,
            width=BAR_WIDTH,
            color=LMUL_COLORS[lmul],
            edgecolor="#222222",
            linewidth=0.38,
            yerr=errs,
            capsize=CAPSIZE,
            error_kw=dict(
                elinewidth=0.70,
                ecolor="#444444",
                capthick=0.70,
            ),
            zorder=3,
        )

        for i, u in enumerate(UNROLLS):
            bar_info[(lmul, u)] = {
                "x": xpos[i],
                "mean": vals[i],
                "std": errs[i],
            }

    for u in UNROLLS:
        records_u = [bar_info[(lmul, u)] for lmul in lmul]
        best_rec = max(records_u, key=lambda r: r["mean"])

        add_label(
            ax=ax,
            x=best_rec["x"],
            y=best_rec["mean"] + best_rec["std"] + LABEL_Y_OFFSET[fam],
            text=VALUE_FORMAT[fam].format(best_rec["mean"]),
            fontsize=LABEL_FONT[fam],
        )

    ax.set_title(
        fam,
        fontsize=10.0,
        fontweight="bold",
        color=FAMILY_ACCENT[fam],
        pad=14,
    )

    ax.text(
        0.5,
        1.016,
        FAMILY_SUBTITLE[fam],
        transform=ax.transAxes,
        ha="center",
        va="bottom",
        fontsize=8.0,
    )

```

```

        fontweight="bold",
        color="#333333",
    )

    ax.set_xticks(x)
    ax.set_xticklabels(
        [UNROLL_LABEL[u] for u in UNROLLS],
        fontsize=9.0,
        fontweight="bold",
    )

    ax.set_xlabel("Unroll factor", fontsize=9.2, labelpad=4)
    ax.set_ylabel(
        "Throughput (GOPS)",
        fontsize=9.6,
        fontweight="bold",
        labelpad=5,
    )

    ax.set_ylim(*Y_LIMITS[fam])
    ax.set_yticks(Y_TICKS[fam])

    ax.yaxis.grid(
        True,
        linestyle=":",
        linewidth=0.50,
        color="#C7C7C7",
        alpha=0.60,
        zorder=0,
    )
    ax.grid(False, axis="x")

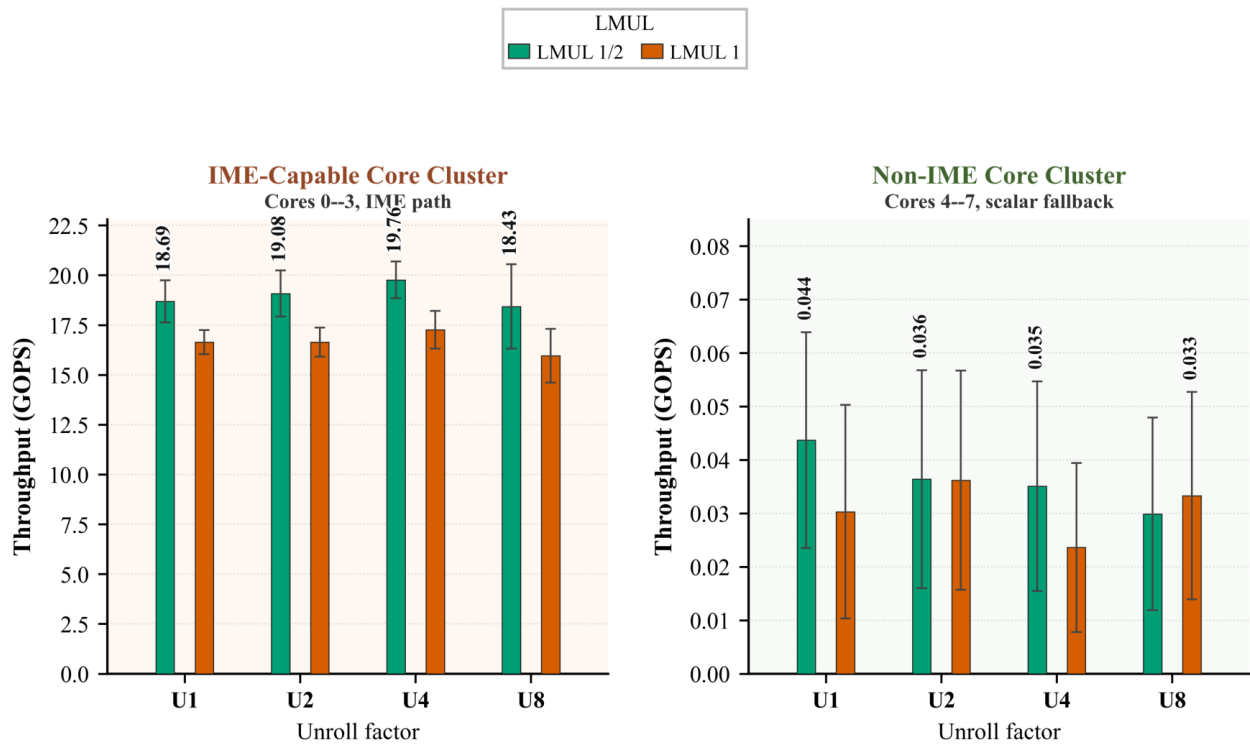
    ax.set_axisbelow(True)
    ax.tick_params(axis="y", labelsize=9.0)
    ax.margins(x=0.12)

legend_handles = [
    Patch(
        facecolor=LMUL_COLORS[l],
        edgecolor="#333333",
        linewidth=0.50,
        label=LMUL_LABEL[l],
    )
    for l in LMUL_ORDER
]

fig.legend(
    handles=legend_handles,
    ncol=2,
    loc="upper center",
    bbox_to_anchor
... [output truncated for PDF readability]
PNG: C:\Users\HP 850 G7\output\ime_acceleration_scalar_fallback_8x4_1024.png
PDF: C:\Users\HP 850 G7\output\ime_acceleration_scalar_fallback_8x4_1024.pdf

```

INT8 GEMM: IME Acceleration and Scalar Fallback 1024³ 8×4 Kernel Across Heterogeneous RISC-V Cores



<Figure size 1950x1200 with 1 Axes>

Code cell 4

```
import os
import numpy as np
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import matplotlib.patheffects as pe
import seaborn as sns
from matplotlib.patches import Patch
from IPython.display import Image, display

# #####
# Springer / publication style
# #####
sns.set_theme(style="whitegrid", context="paper")

matplotlib.rcParams.update({
    "pdf.fonttype": 42,
    "ps.fonttype": 42,
    "font.family": "serif",
    "font.serif": ["Times New Roman", "Times", "DejaVu Serif"],
    "mathtext.fontset": "stix",

    "axes.linewidth": 0.90,
    "axes.edgecolor": "black",
    "axes.spines.top": False,
    "axes.spines.right": False,

    "xtick.major.width": 0.80,
    "ytick.major.width": 0.80,
    "xtick.major.size": 3.6,
    "ytick.major.size": 3.6,
    "xtick.direction": "out",
    "ytick.direction": "out",
})

# #####
# Output
# #####
```



```

    "mf4": "LMUL 1/4",
    "mf2": "LMUL 1/2",
    "1": "LMUL 1",
    "2": "LMUL 2",
    "4": "LMUL 4",
    "8": "LMUL 8",
}

palette = sns.color_palette("colorblind", 7)

LMUL_COLORS = {
    "mf8": palette[0],
    "mf4": palette[1],
    "mf2": palette[2],
    "1": palette[3],
    "2": palette[4],
    "4": palette[5],
    "8": palette[6],
}

FAMILY_FACE = {
    "FP64": "#F7F9FC",
    "INT8": "#F7FAF6",
    "IME": "#FFF8F2",
}

FAMILY_ACCENT = {
    "FP64": "#1B3A5C",
    "INT8": "#2D4A1E",
    "IME": "#7A2A1E",
}

FAMILY_SUBTITLE = {
    "FP64": r"FP64$\times$FP64$\rightarrow$FP64",
    "INT8": r"INT8$\times$INT8$\rightarrow$INT32",
    "IME": r"INT8$\times$INT8$\rightarrow$INT32",
}

FAMILY_YLABEL = {
    "FP64": "GFLOPS",
    "INT8": "GOPS",
    "IME": "GOPS",
}

UNROLL_LABEL = {
    1: "U1",
    2: "U2",
    4: "U4",
    8: "U8",
}

# Polished headroom for large labels
Y_LIMITS = {
    "FP64": (0.0, 5.85),
    "INT8": (0.0, 13.10),
    "IME": (19.0, 20.80),
}

BAR_WIDTH = {
    "FP64": 0.115,
    "INT8": 0.078,
    "IME": 0.125,
}

GROUP_WIDTH = {
    "FP64": 0.56,
    "INT8": 0.74,
    "IME": 0.32,
}

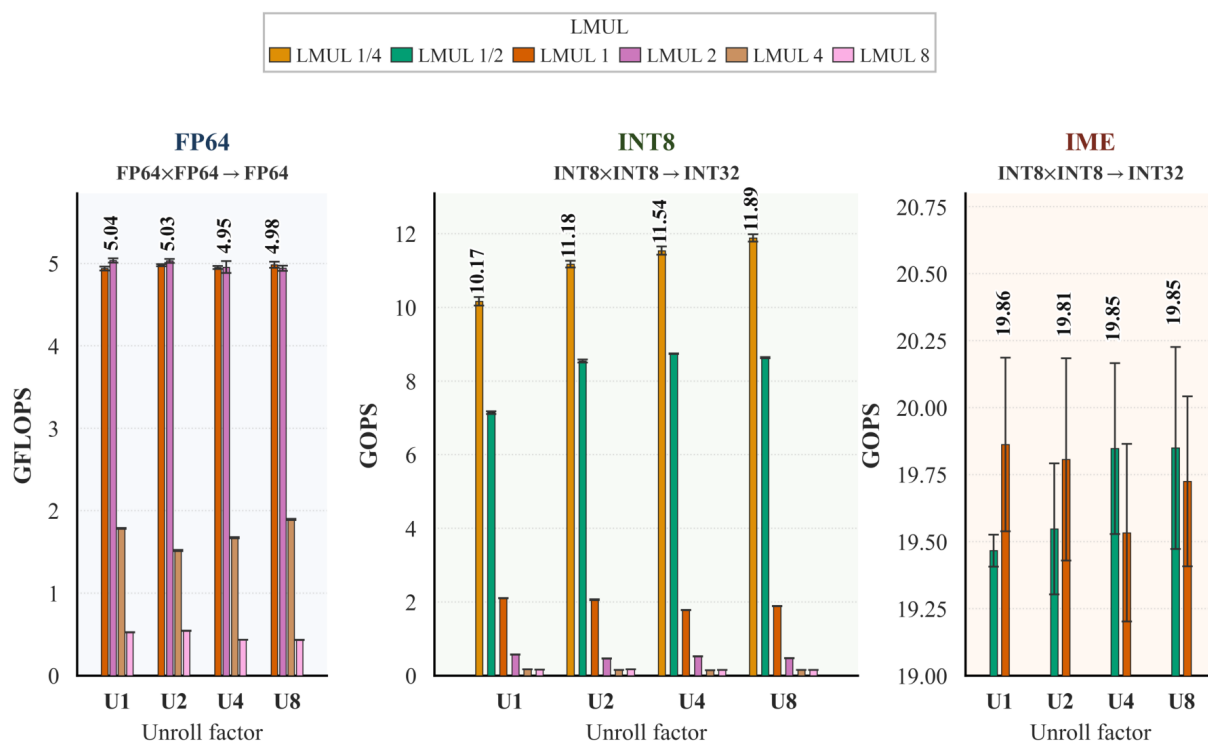
LABEL_FONT = {
    "FP64": 8.0,
    "INT8": 8.0,
    "IME": 8.0,
}

CAPSIZE = 2.0

LABEL_Y_OFFSET = {
    "FP64": 0.090,
    "INT8": 0.080,
    "IME": 0.080,
}

```


Throughput Tuning of FP64 OpenBLAS 8x8, INT8, and IME for 1024³ GEMM



<Figure size 2100x1275 with 1 Axes>

Code cell 5

```
import os
import numpy as np
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import matplotlib.patheffects as pe
from matplotlib.patches import Patch

plt.close("all")

matplotlib.rcParams.update({
    "pdf.fonttype": 42,
    "ps.fonttype": 42,
    "font.family": "serif",
    "font.serif": ["Times New Roman", "Times", "DejaVu Serif"],
    "mathtext.fontset": "stix",
    "axes.linewidth": 0.90,
    "xtick.major.width": 0.80,
    "ytick.major.width": 0.80,
    "xtick.minor.width": 0.60,
    "ytick.minor.width": 0.60,
    "xtick.major.size": 3.6,
    "ytick.major.size": 3.6,
    "xtick.direction": "out",
    "ytick.direction": "out",
    "axes.spines.top": False,
    "axes.spines.right": False,
    "figure.facecolor": "white",
    "savefig.facecolor": "white",
})

OUTPUT_DIR = "output"
os.makedirs(OUTPUT_DIR, exist_ok=True)

SAVE_PATH_PNG = os.path.join(
    OUTPUT_DIR,
    "ime_acceleration_scalar_fallback_8x8_1024.png"
)
```

```

SAVE_PATH_PDF = os.path.join(
    OUTPUT_DIR,
    "ime_acceleration_scalar_fallback_8x8_1024.pdf"
)

FIG_W, FIG_H = 6.8, 4.65
BAR_WIDTH = 0.16
GROUP_WIDTH = 0.34
CAPSIZE = 2.0

DATA = [
    ("IME-Capable Core Cluster", "1", 1, 19.86201666666667, 0.324074186037018),
    ("IME-Capable Core Cluster", "1", 2, 19.80710000000000, 0.377561169437042),
    ("IME-Capable Core Cluster", "1", 4, 19.53326666666667, 0.331679316341686),
    ("IME-Capable Core Cluster", "1", 8, 19.72445000000000, 0.317294278496582),

    ("IME-Capable Core Cluster", "mf2", 1, 19.46671666666667, 0.059715392115900),
    ("IME-Capable Core Cluster", "mf2", 2, 19.54775000000000, 0.244229733038383),
    ("IME-Capable Core Cluster", "mf2", 4, 19.84740000000000, 0.319252475845268),
    ("IME-Capable Core Cluster", "mf2", 8, 19.84991666666667, 0.376999546492507),

    ("Non-IME Core Cluster", "1", 1, 1.024500000000000, 0.001678292783357),
    ("Non-IME Core Cluster", "1", 2, 1.025500000000000, 0.001561783168902),
    ("Non-IME Core Cluster", "1", 4, 1.025733333333333, 0.001167142760001),
    ("Non-IME Core Cluster", "1", 8, 1.026033333333333, 0.000975818744554),

    ("Non-IME Core Cluster", "mf2", 1, 1.025416666666667, 0.002489589435139),
    ("Non-IME Core Cluster", "mf2", 2, 1.026450000000000, 0.000442530601578),
    ("Non-IME Core Cluster", "mf2", 4, 1.024616666666667, 0.001917825736493),
    ("Non-IME Core Cluster", "mf2", 8, 1.025916666666667, 0.001294110591187),
]

df = pd.DataFrame(DATA, columns=["Family", "LMUL", "Unroll", "Mean", "Std"])

FAMILIES = [
    "IME-Capable Core Cluster",
    "Non-IME Core Cluster",
]

UNROLLS = [1, 2, 4, 8]
LMUL_ORDER = ["mf2", "1"]

LMUL_LABEL = {
    "mf2": "LMUL 1/2",
    "1": "LMUL 1",
}

LMUL_COLORS = {
    "mf2": "#009E73",
    "1": "#D55E00",
}

FAMILY_FACE = {
    "IME-Capable Core Cluster": "#FFF8F2",
    "Non-IME Core Cluster": "#F7FAF6",
}

FAMILY_ACCENT = {
    "IME-Capable Core Cluster": "#944727",
    "Non-IME Core Cluster": "#41642F",
}

FAMILY_SUBTITLE = {
    "IME-Capable Core Cluster": "Cores 0--3, IME path",
    "Non-IME Core Cluster": "Cores 4--7, scalar fallback",
}

UNROLL_LABEL = {
    1: "U1",
    2: "U2",
    4: "U4",
    8: "U8",
}

Y_LIMITS = {
    "IME-Capable Core Cluster": (19.0, 20.80),
    "Non-IME Core Cluster": (0.0, 1.16),
}

Y_TICKS = {
    "IME-Capable Core Cluster": np.arange(19.0, 20.81, 0.4),
    "Non-IME Core Cluster": np.arange(0.0, 1.11, 0.2),
}

```

```

}

LABEL_FONT = {
    "IME-Capable Core Cluster": 8.0,
    "Non-IME Core Cluster": 7.6,
}

LABEL_Y_OFFSET = {
    "IME-Capable Core Cluster": 0.035,
    "Non-IME Core Cluster": 0.008,
}

VALUE_FORMAT = {
    "IME-Capable Core Cluster": "{:.2f}",
    "Non-IME Core Cluster": "{:.3f}",
}

def add_label(ax, x, y, text, fontsize):
    label = ax.text(
        x,
        y,
        text,
        ha="center",
        va="bottom",
        rotation=90,
        fontsize=fontsize,
        fontweight="bold",
        color="#111111",
        zorder=8,
        clip_on=False,
    )
    label.set_path_effects([
        pe.Stroke(linewidth=1.60, foreground="white"),
        pe.Normal(),
    ])
    return label

fig, axes = plt.subplots(
    1, 2,
    figsize=(FIG_W, FIG_H),
    dpi=300,
    constrained_layout=False,
    gridspec_kw={
        "wspace": 0.28,
        "width_ratios": [1, 1],
    },
)

fig.patch.set_facecolor("white")

fig.suptitle(
    r"INT8 GEMM: IME Acceleration and Scalar Fallback"
    "\n"
    r"$1024^3$ 8$\times$8 Kernel Across Heterogeneous RISC-V Cores",
    fontsize=10.0,
    fontweight="bold",
    y=0.975,
)

for ax, fam in zip(axes, FAMILIES):
    sub = df[df["Family"] == fam].copy()
    lmul = [l for l in LMUL_ORDER if l in set(sub["LMUL"])]

    x = np.arange(len(UNROLLS))
    offsets = np.linspace(-GROUP_WIDTH / 2, GROUP_WIDTH / 2, len(lmul))

    ax.set_facecolor(FAMILY_FACE[fam])
    bar_info = {}

    for offset, lmul in zip(offsets, lmul):
        s = sub[sub["LMUL"] == lmul].set_index("Unroll").reindex(UNROLLS)
        vals = s["Mean"].values
        errs = s["Std"].values
        xpos = x + offset

        ax.bar(
            xpos,
            vals,
            width=BAR_WIDTH,
            color=LMUL_COLORS[lmul],

```

```

        edgecolor="#222222",
        linewidth=0.38,
        yerr=errs,
        capsize=CAPSIZE,
        error_kw=dict(
            elinewidth=0.70,
            ecolor="#444444",
            capthick=0.70,
        ),
        zorder=3,
    )

    for i, u in enumerate(UNROLLS):
        bar_info[(lmul, u)] = {
            "x": xpos[i],
            "mean": vals[i],
            "std": errs[i],
        }

for u in UNROLLS:
    records_u = [bar_info[(lmul, u)] for lmul in lmul]
    best_rec = max(records_u, key=lambda r: r["mean"])

    add_label(
        ax=ax,
        x=best_rec["x"],
        y=best_rec["mean"] + best_rec["std"] + LABEL_Y_OFFSET[fam],
        text=VALUE_FORMAT[fam].format(best_rec["mean"]),
        fontsize=LABEL_FONT[fam],
    )

ax.set_title(
    fam,
    fontsize=10.0,
    fontweight="bold",
    color=FAMILY_ACCENT[fam],
    pad=14,
)

ax.text(
    0.5,
    1.016,
    FAMILY_SUBTITLE[fam],
    transform=ax.transAxes,
    ha="center",
    va="bottom",
    fontsize=8.0,
    fontweight="bold",
    color="#333333",
)

ax.set_xticks(x)
ax.set_xticklabels(
    [UNROLL_LABEL[u] for u in UNROLLS],
    fontsize=9.0,
    fontweight="bold",
)

ax.set_xlabel("Unroll factor", fontsize=9.2, labelpad=4)
ax.set_ylabel(
    "Throughput (GOPS)",
    fontsize=9.6,
    fontweight="bold",
    labelpad=5,
)

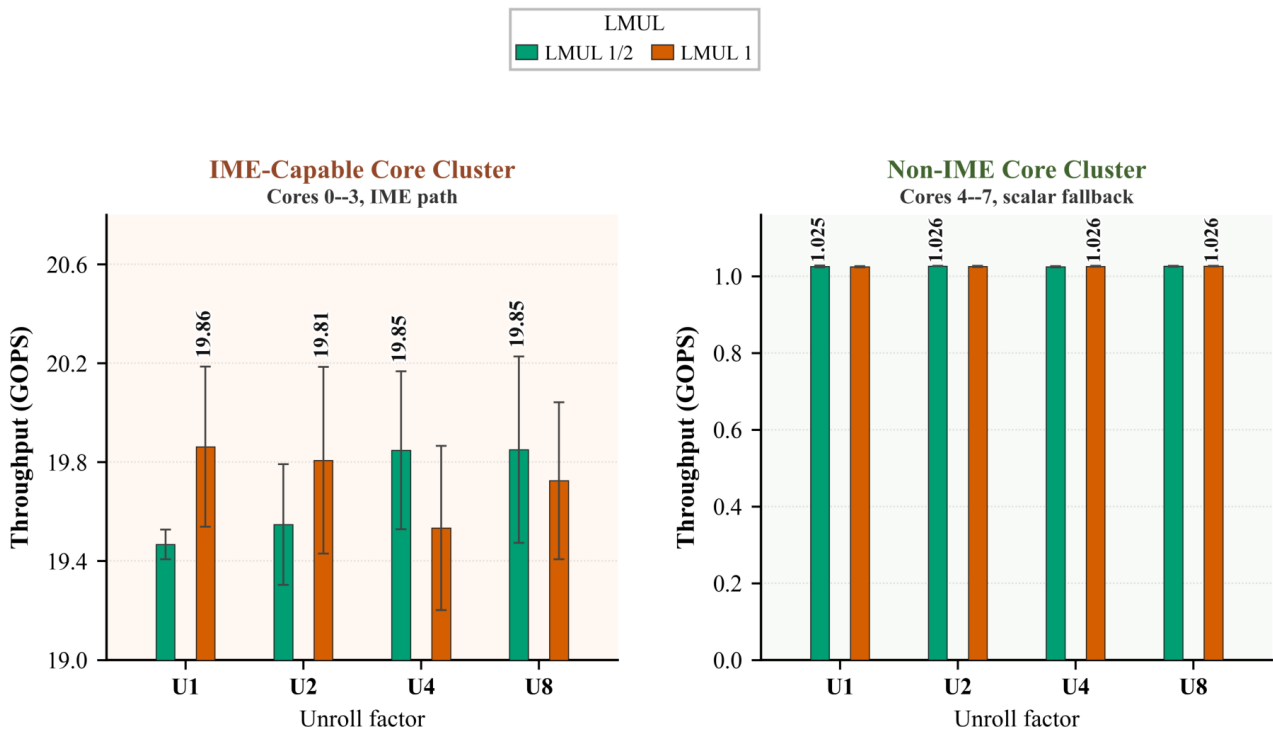
ax.set_ylim(*Y_LIMITS[fam])
ax.set_yticks(Y_TICKS[fam])

ax.yaxis.grid(
    True,
    linestyle=":",
    linewidth=0.50,
    color="#C7C7C7",
    alpha=0.60,
    zorder=0,
)
ax.grid(False, axis="x")

ax.set_axisbelow(True)
ax
... [output truncated for PDF readability]

```

INT8 GEMM: IME Acceleration and Scalar Fallback 1024³ 8x8 Kernel Across Heterogeneous RISC-V Cores



<Figure size 1950x1200 with 1 Axes>

Code cell 6

```
import os
import numpy as np
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import matplotlib.patheffects as pe
import seaborn as sns
from matplotlib.patches import Patch
from IPython.display import Image, display

# #####
# Springer / publication style
# #####
sns.set_theme(style="whitegrid", context="paper")

matplotlib.rcParams.update({
    "pdf.fonttype": 42,
    "ps.fonttype": 42,
    "font.family": "serif",
    "font.serif": ["Times New Roman", "Times", "DejaVu Serif"],
    "mathtext.fontset": "stix",

    "axes.linewidth": 0.90,
    "axes.edgecolor": "black",
    "axes.spines.top": False,
    "axes.spines.right": False,

    "xtick.major.width": 0.80,
    "ytick.major.width": 0.80,
    "xtick.major.size": 3.6,
    "ytick.major.size": 3.6,
    "xtick.direction": "out",
    "ytick.direction": "out",
})

# #####
```



```
df = pd.DataFrame(DATA, columns=["Family", "LMUL", "Unroll", "Mean", "Std"])

# #####
# Constants
# #####

FAMILIES = ["FP32", "INT8", "IME"]
UNROLLS = [1, 2, 4, 8]
LMUL_ORDER = ["mf8", "mf4", "mf2", "1", "2", "4", "8"]

LMUL_LABEL = {
    "mf8": "LMUL 1/8",
    "mf4": "LMUL 1/4",
    "mf2": "LMUL 1/2",
    "1": "LMUL 1",
    "2": "LMUL 2",
    "4": "LMUL 4",
    "8": "LMUL 8",
}

palette = sns.color_palette("colorblind", 7)

LMUL_COLORS = {
    "mf8": palette[0],
    "mf4": palette[1],
    "mf2": palette[2],
    "1": palette[3],
    "2": palette[4],
    "4": palette[5],
    "8": palette[6],
}

FAMILY_FACE = {
    "FP32": "#F7F9FC",
    "INT8": "#F7FAF6",
    "IME": "#FFF8F2",
}

FAMILY_ACCENT = {
    "FP32": "#1B3A5C",
    "INT8": "#2D4A1E",
    "IME": "#7A2A1E",
}

FAMILY_SUBTITLE = {
    "FP32": r"FP32$\times$FP32$\rightarrow$FP32",
    "INT8": r"INT8$\times$INT8$\rightarrow$INT32",
    "IME": r"INT8$\times$INT8$\rightarrow$INT32",
}

FAMILY_YLABEL = {
    "FP32": "GFLOPS",
    "INT8": "GOPs",
    "IME": "GOPs",
}

UNROLL_LABEL = {
    1: "U1",
    2: "U2",
    4: "U4",
    8: "U8",
}

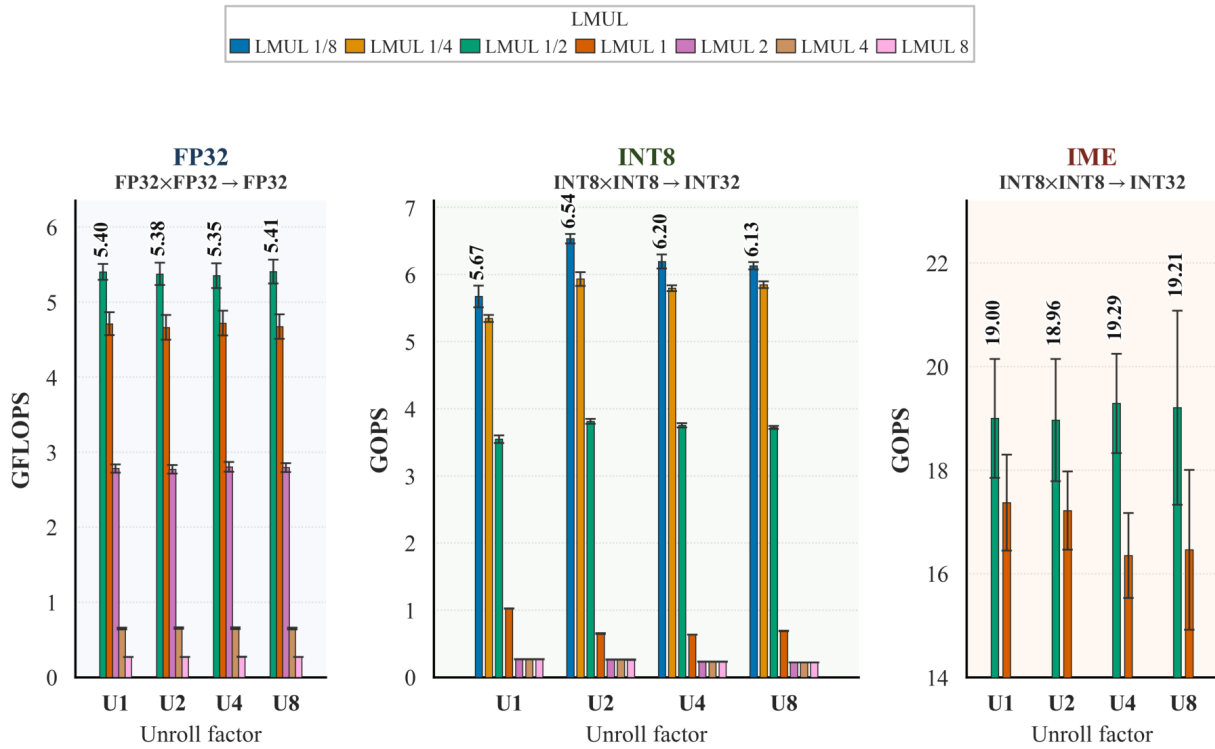
Y_LIMITS = {
    "FP32": (0.0, 6.35),
    "INT8": (0.0, 7.10),
    "IME": (14.0, 23.20),
}

BAR_WIDTH = {
    "FP32": 0.115,
    "INT8": 0.078,
    "IME": 0.125,
}

GROUP_WIDTH = {
    "FP32": 0.56,
    "INT8": 0.74,
    "IME": 0.32,
}

LABEL_FONT = {
    "FP32": 8.0,
```


Throughput Tuning of FP32 OpenBLAS 8×4, INT8, and IME for 1024³ GEMM



<Figure size 2100x1275 with 1 Axes>

Code cell 7

```
import os
import numpy as np
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.image as mimg
import matplotlib.patheffects as pe
from matplotlib.patches import Patch

plt.close("all")

matplotlib.rcParams.update({
    "pdf.fonttype": 42,
    "ps.fonttype": 42,
    "font.family": "serif",
    "font.serif": ["Times New Roman", "Times", "DejaVu Serif"],
    "mathtext.fontset": "stix",
    "axes.linewidth": 0.90,
    "xtick.major.width": 0.80,
    "ytick.major.width": 0.80,
    "xtick.minor.width": 0.60,
    "ytick.minor.width": 0.60,
    "xtick.major.size": 3.6,
    "ytick.major.size": 3.6,
    "xtick.direction": "out",
    "ytick.direction": "out",
    "axes.spines.top": False,
    "axes.spines.right": False,
    "figure.facecolor": "white",
    "savefig.facecolor": "white",
})

OUTPUT_DIR = "output"
os.makedirs(OUTPUT_DIR, exist_ok=True)

SAVE_PATH_PNG = os.path.join(
    OUTPUT_DIR,
    "ime_acceleration_scalar_fallback_fp32_8x4_1024.png"
)
```

```

SAVE_PATH_PDF = os.path.join(
    OUTPUT_DIR,
    "ime_acceleration_scalar_fallback_fp32_8x4_1024.pdf"
)

FIG_W, FIG_H = 6.8, 4.65
BAR_WIDTH = 0.16
GROUP_WIDTH = 0.34
CAPSIZE = 2.0

DATA = [
    ("IME-Capable Core Cluster", "1", 1, 17.373050000000, 0.925107978833),
    ("IME-Capable Core Cluster", "1", 2, 17.217466666667, 0.756900587190),
    ("IME-Capable Core Cluster", "1", 4, 16.352000000000, 0.818897516991),
    ("IME-Capable Core Cluster", "1", 8, 16.461900000000, 1.542673804363),

    ("IME-Capable Core Cluster", "mf2", 1, 18.997733333333, 1.149666336619),
    ("IME-Capable Core Cluster", "mf2", 2, 18.963850000000, 1.179027563644),
    ("IME-Capable Core Cluster", "mf2", 4, 19.287100000000, 0.959837672040),
    ("IME-Capable Core Cluster", "mf2", 8, 19.205483333333, 1.872285032980),

    ("Non-IME Core Cluster", "1", 1, 0.029966666667, 0.019533276450),
    ("Non-IME Core Cluster", "1", 2, 0.030583333333, 0.017666674528),
    ("Non-IME Core Cluster", "1", 4, 0.042550000000, 0.017232116334),
    ("Non-IME Core Cluster", "1", 8, 0.035983333333, 0.018701373984),

    ("Non-IME Core Cluster", "mf2", 1, 0.016183333333, 0.001591033906),
    ("Non-IME Core Cluster", "mf2", 2, 0.036583333333, 0.019404330845),
    ("Non-IME Core Cluster", "mf2", 4, 0.042250000000, 0.020076250480),
    ("Non-IME Core Cluster", "mf2", 8, 0.049200000000, 0.014336550027),
]

df = pd.DataFrame(DATA, columns=["Family", "LMUL", "Unroll", "Mean", "Std"])

FAMILIES = [
    "IME-Capable Core Cluster",
    "Non-IME Core Cluster",
]

UNROLLS = [1, 2, 4, 8]
LMUL_ORDER = ["mf2", "1"]

LMUL_LABEL = {
    "mf2": "LMUL 1/2",
    "1": "LMUL 1",
}

LMUL_COLORS = {
    "mf2": "#009E73",
    "1": "#D55E00",
}

FAMILY_FACE = {
    "IME-Capable Core Cluster": "#FFF8F2",
    "Non-IME Core Cluster": "#F7FAF6",
}

FAMILY_ACCENT = {
    "IME-Capable Core Cluster": "#944727",
    "Non-IME Core Cluster": "#41642F",
}

FAMILY_SUBTITLE = {
    "IME-Capable Core Cluster": "Cores 0--3, IME path",
    "Non-IME Core Cluster": "Cores 4--7, scalar fallback",
}

UNROLL_LABEL = {
    1: "U1",
    2: "U2",
    4: "U4",
    8: "U8",
}

Y_LIMITS = {
    "IME-Capable Core Cluster": (15.0, 22.5),
    "Non-IME Core Cluster": (0.0, 0.080),
}

Y_TICKS = {
    "IME-Capable Core Cluster": np.arange(15.0, 22.6, 1.5),
    "Non-IME Core Cluster": np.arange(0.0, 0.081, 0.01),
}

```

```

}

LABEL_FONT = {
    "IME-Capable Core Cluster": 8.0,
    "Non-IME Core Cluster": 7.6,
}

LABEL_Y_OFFSET = {
    "IME-Capable Core Cluster": 0.10,
    "Non-IME Core Cluster": 0.0016,
}

VALUE_FORMAT = {
    "IME-Capable Core Cluster": "{:.2f}",
    "Non-IME Core Cluster": "{:.3f}",
}

def add_label(ax, x, y, text, fontsize):
    label = ax.text(
        x,
        y,
        text,
        ha="center",
        va="bottom",
        rotation=90,
        fontsize=fontsize,
        fontweight="bold",
        color="#111111",
        zorder=8,
        clip_on=False,
    )
    label.set_path_effects([
        pe.Stroke(linewidth=1.60, foreground="white"),
        pe.Normal(),
    ])
    return label

fig, axes = plt.subplots(
    1, 2,
    figsize=(FIG_W, FIG_H),
    dpi=300,
    constrained_layout=False,
    gridspec_kw={"wspace": 0.28, "width_ratios": [1, 1]},
)

fig.patch.set_facecolor("white")

fig.suptitle(
    r"INT8 GEMM: IME Acceleration and Scalar Fallback"
    "\n"
    r"$1024^3$ 8$ \times $4 Kernel Across Heterogeneous RISC-V Cores",
    fontsize=10.0,
    fontweight="bold",
    y=0.975,
)

for ax, fam in zip(axes, FAMILIES):
    sub = df[df["Family"] == fam].copy()
    lmul = [l for l in LMUL_ORDER if l in set(sub["LMUL"])]

    x = np.arange(len(UNROLLS))
    offsets = np.linspace(-GROUP_WIDTH / 2, GROUP_WIDTH / 2, len(lmul))

    ax.set_facecolor(FAMILY_FACE[fam])
    bar_info = {}

    for offset, lmul in zip(offsets, lmul):
        s = sub[sub["LMUL"] == lmul].set_index("Unroll").reindex(UNROLLS)
        vals = s["Mean"].values
        errs = s["Std"].values
        xpos = x + offset

        ax.bar(
            xpos,
            vals,
            width=BAR_WIDTH,
            color=LMUL_COLORS[lmul],
            edgecolor="#222222",
            linewidth=0.38,
            yerr=errs,

```

```

        capsize=CAPSIZE,
        error_kw=dict(elinewidth=0.70, ecolord="#444444", capthick=0.70),
        zorder=3,
    )

    for i, u in enumerate(UNROLLS):
        bar_info[(lmul, u)] = {
            "x": xpos[i],
            "mean": vals[i],
            "std": errs[i],
        }

    for u in UNROLLS:
        records_u = [bar_info[(lmul, u)] for lmul in lmul]
        best_rec = max(records_u, key=lambda r: r["mean"])

        add_label(
            ax=ax,
            x=best_rec["x"],
            y=best_rec["mean"] + best_rec["std"] + LABEL_Y_OFFSET[fam],
            text=VALUE_FORMAT[fam].format(best_rec["mean"]),
            fontsize=LABEL_FONT[fam],
        )

    ax.set_title(
        fam,
        fontsize=10.0,
        fontweight="bold",
        color=FAMILY_ACCENT[fam],
        pad=14,
    )

    ax.text(
        0.5,
        1.016,
        FAMILY_SUBTITLE[fam],
        transform=ax.transAxes,
        ha="center",
        va="bottom",
        fontsize=8.0,
        fontweight="bold",
        color="#333333",
    )

    ax.set_xticks(x)
    ax.set_xticklabels(
        [UNROLL_LABEL[u] for u in UNROLLS],
        fontsize=9.0,
        fontweight="bold",
    )

    ax.set_xlabel("Unroll factor", fontsize=9.2, labelpad=4)
    ax.set_ylabel(
        "Throughput (GOPS)",
        fontsize=9.6,
        fontweight="bold",
        labelpad=5,
    )

    ax.set_ylim(*Y_LIMITS[fam])
    ax.set_yticks(Y_TICKS[fam])

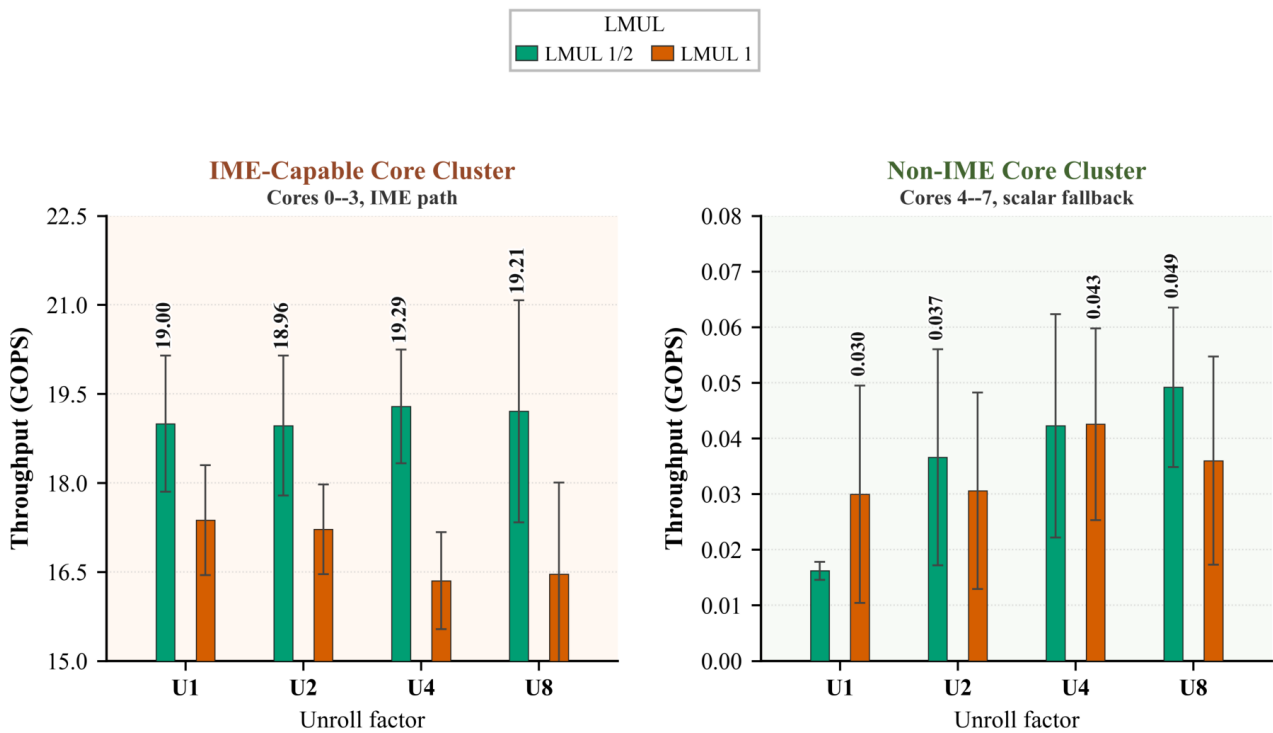
    ax.yaxis.grid(
        True,
        linestyle=":",
        linewidth=0.50,
        color="#C7C7C7",
        alpha=0.60,
        zorder=0,
    )
    ax.grid(False, axis="x")
    ax.set_axisbelow(True)
    ax.tick_params(axis="y", labelsize=9.0)
    ax.margins(x=0.12)

    legend_handles = [
        Patch(
            facecolor=LMUL_COLORS[l],
            edgecolor="#33
... [output truncated for PDF readability]

```

PNG: C:\Users\HP 850 G7\output\ime_acceleration_scalar_fallback_fp32_8x4_1024.png

INT8 GEMM: IME Acceleration and Scalar Fallback 1024³ 8×4 Kernel Across Heterogeneous RISC-V Cores



<Figure size 1950x1200 with 1 Axes>

Code cell 8

```
import os
import numpy as np
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.image as mimg
import matplotlib.patheffects as pe
import seaborn as sns
from matplotlib.patches import Patch
from IPython.display import Image, display

# #####
# Springer / publication style
# #####
sns.set_theme(style="whitegrid", context="paper")

matplotlib.rcParams.update({
    "pdf.fonttype": 42,
    "ps.fonttype": 42,
    "font.family": "serif",
    "font.serif": ["Times New Roman", "Times", "DejaVu Serif"],
    "mathtext.fontset": "stix",

    "axes.linewidth": 0.90,
    "axes.edgecolor": "black",
    "axes.spines.top": False,
    "axes.spines.right": False,

    "xtick.major.width": 0.80,
    "ytick.major.width": 0.80,
    "xtick.major.size": 3.6,
    "ytick.major.size": 3.6,
    "xtick.direction": "out",
    "ytick.direction": "out",
})

# #####
# Output
```



```

    "mf8": "LMUL 1/8",
    "mf4": "LMUL 1/4",
    "mf2": "LMUL 1/2",
    "1": "LMUL 1",
    "2": "LMUL 2",
    "4": "LMUL 4",
    "8": "LMUL 8",
}

palette = sns.color_palette("colorblind", 7)

LMUL_COLORS = {
    "mf8": palette[0],
    "mf4": palette[1],
    "mf2": palette[2],
    "1": palette[3],
    "2": palette[4],
    "4": palette[5],
    "8": palette[6],
}

FAMILY_FACE = {
    "FP32": "#F7F9FC",
    "INT8": "#F7FAF6",
    "IME": "#FFF8F2",
}

FAMILY_ACCENT = {
    "FP32": "#1B3A5C",
    "INT8": "#2D4A1E",
    "IME": "#7A2A1E",
}

FAMILY_SUBTITLE = {
    "FP32": r"FP32$\times$FP32$\rightarrow$FP32",
    "INT8": r"INT8$\times$INT8$\rightarrow$INT32",
    "IME": r"INT8$\times$INT8$\rightarrow$INT32",
}

FAMILY_YLABEL = {
    "FP32": "GFLOPS",
    "INT8": "GOPs",
    "IME": "GOPs",
}

UNROLL_LABEL = {
    1: "U1",
    2: "U2",
    4: "U4",
    8: "U8",
}

Y_LIMITS = {
    "FP32": (0.0, 9.2),
    "INT8": (0.0, 12.8),
    "IME": (18.8, 20.9),
}

BAR_WIDTH = {
    "FP32": 0.115,
    "INT8": 0.078,
    "IME": 0.125,
}

GROUP_WIDTH = {
    "FP32": 0.56,
    "INT8": 0.74,
    "IME": 0.32,
}

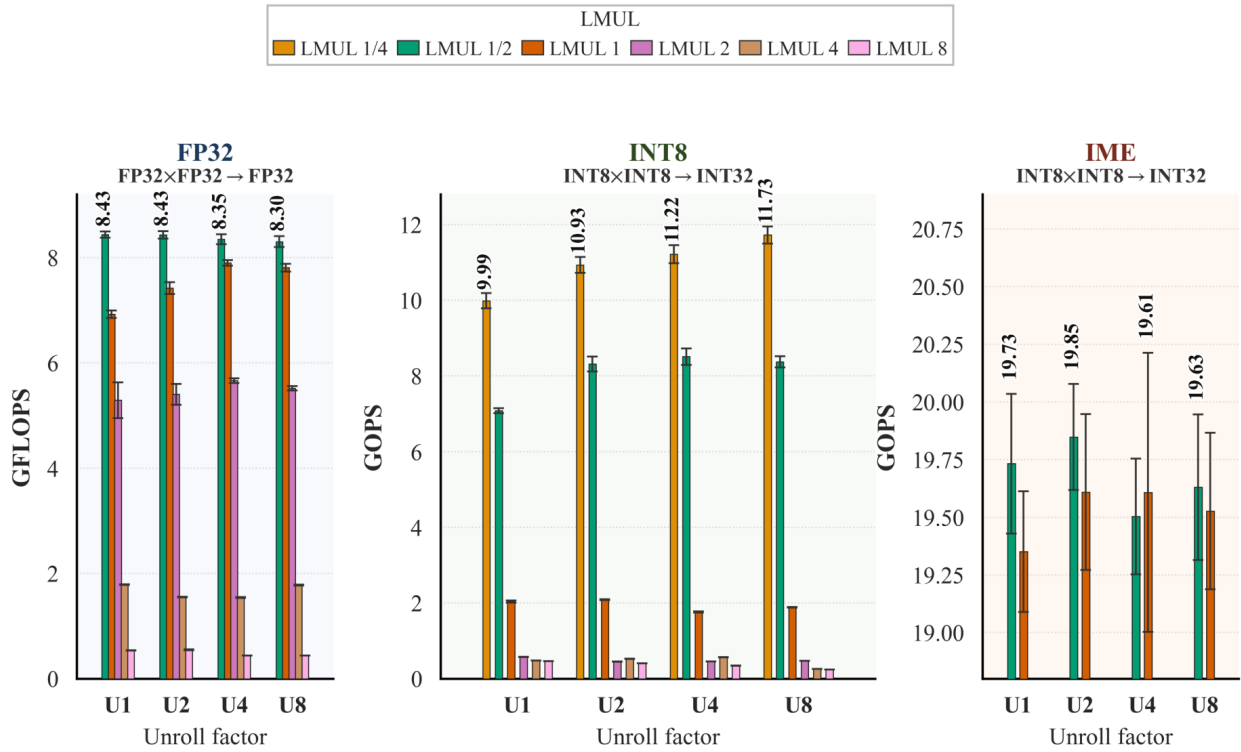
LABEL_FONT = {
    "FP32": 8.0,
    "INT8": 8.0,
    "IME": 8.0,
}

CAPSIZE = 2.0

LABEL_Y_OFFSET = {
    "FP32": 0.090,
    "INT8": 0.080,
    "IME": 0.060,
}

```


Throughput Tuning of FP32 OpenBLAS 8×8, INT8, and IME for 1024³ GEMM



<Figure size 2100x1275 with 1 Axes>

Code cell 9

```
import os
import numpy as np
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
import matplotlib.image as mimg
import matplotlib.path as mpath
from matplotlib.patches import Patch
```

```
# CLEAN START
plt.close("all")

# Springer / publication style – polished large-font version
matplotlib.rcParams.update({
    "pdf.fonttype": 42,
    "ps.fonttype": 42,
    "font.family": "serif",
    "font.serif": ["Times New Roman", "Times", "DejaVu Serif"],
    "mathtext.fontset": "stix",

    "axes.linewidth": 0.90,
    "xtick.major.width": 0.80,
    "ytick.major.width": 0.80,
    "xtick.minor.width": 0.60,
    "ytick.minor.width": 0.60,
    "xtick.major.size": 3.6,
    "ytick.major.size": 3.6,
    "xtick.direction": "out",
    "ytick.direction": "out",
    "axes.spines.top": False,
    "axes.spines.right": False,
    "figure.facecolor": "white",
    "savefig.facecolor": "white",
})
```

```
#
# CONFIG
#
OUTPUT_DIR = "output"
os.makedirs(OUTPUT_DIR, exist_ok=True)

SAVE_PATH_PNG = os.path.join(
    OUTPUT_DIR,
    "ime_core_cluster_throughput_8x8_springer_polished_fp32config_2line.png"
)
SAVE_PATH_PDF = os.path.join(
    OUTPUT_DIR,
    "ime_core_cluster_throughput_8x8_springer_polished_fp32config_2line.pdf"
)

FIG_W, FIG_H = 6.8, 4.65
BAR_WIDTH = 0.16
GROUP_WIDTH = 0.34
CAPSIZE = 2.0

#
# DATA
# Fresh FP32-INT-IME 8x8 baseline summary_results_1024
# IME family separated by core clusters
#
DATA = [
    # IME(RVV) AI Core Cluster, cores 0-3
    ("IME(RVV) AI Core Cluster", "1", 1, 19.351283333333, 0.261820596444),
    ("IME(RVV) AI Core Cluster", "1", 2, 19.609300000000, 0.338378077501),
    ("IME(RVV) AI Core Cluster", "1", 4, 19.607716666667, 0.605230750531),
    ("IME(RVV) AI Core Cluster", "1", 8, 19.526866666667, 0.338955285481),

    ("IME(RVV) AI Core Cluster", "mf2", 1, 19.732116666667, 0.302594356726),
    ("IME(RVV) AI Core Cluster", "mf2", 2, 19.847650000000, 0.230171289478),
    ("IME(RVV) AI Core Cluster", "mf2", 4, 19.503700000000, 0.251125917951),
    ("IME(RVV) AI Core Cluster", "mf2", 8, 19.630216666667, 0.315945941667),

    # General(RVV) Core Cluster, cores 4-6
    ("General(RVV) Core Cluster", "1", 1, 1.026050000000, 0.001480709289),
    ("General(RVV) Core Cluster", "1", 2, 1.026133333333, 0.001308519095),
    ("General(RVV) Core Cluster", "1", 4, 1.026366666667, 0.000859586464),
    ("General(RVV) Core Cluster", "1", 8, 1.025133333333, 0.001482865095),

    ("General(RVV) Core Cluster", "mf2", 1, 1.022766666667, 0.005837141614),
    ("General(RVV) Core Cluster", "mf2", 2, 1.025616666667, 0.001329682502),
    ("General(RVV) Core Cluster", "mf2", 4, 1.025300000000, 0.002059935274),
    ("General(RVV) Core Cluster", "mf2", 8, 1.024000000000, 0.004116632928),
]

df = pd.DataFrame(DATA, columns=["Family", "LMUL", "Unroll", "Mean", "Std"])

#
# CONSTANTS
#
FAMILIES = [
    "IME(RVV) AI Core Cluster",
    "General(RVV) Core Cluster",
]

UNROLLS = [1, 2, 4, 8]
LMUL_ORDER = ["mf2", "1"]

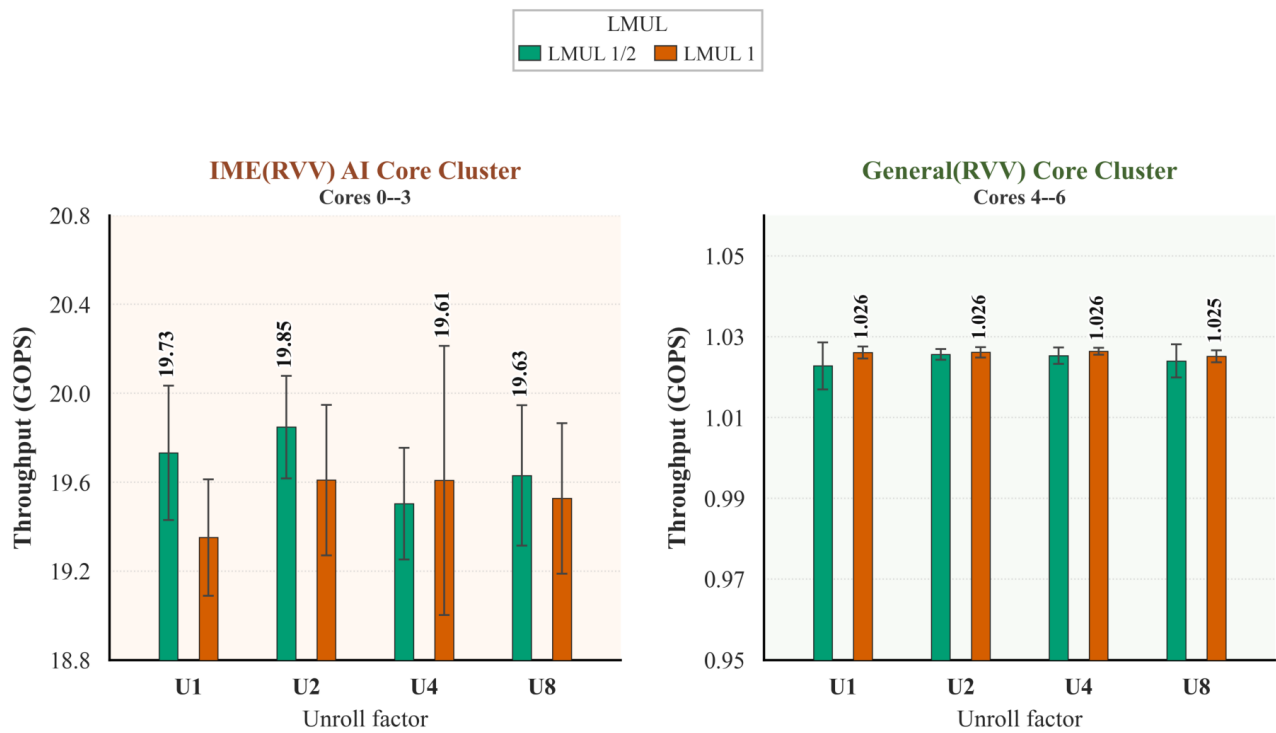
LMUL_LABEL = {
    "mf2": "LMUL 1/2",
    "1": "LMUL 1",
}

LMUL_COLORS = {
    "mf2": "#009E73",
    "1": "#D55E00",
}

FAMILY_FACE = {
    "IME(RVV) AI Core Cluster": "#FFF8F2",
    "General(RVV) Core Cluster": "#F7FAF6",
}

FAMILY_ACCENT = {
    "IME(RVV) AI Core Cluster": "#944727",
    "General(RVV) Core Cluster": "#41642F",
}
```


IME Throughput across Core Clusters
FP32 OpenBLAS 8x8 Configuration for 1024³ GEMM



<Figure size 1950x1200 with 1 Axes>

Code cell 10

Code cell 11

Code cell 12

Code cell 13